WORKSHEET 0.1

Functions You Must Know Cold

Unit 0: Readiness

Key ideas

- A **domain** excludes zero denominators, negative numbers under even roots, and non-positive inputs of logarithms.
- Changes *inside* the brackets act on x and work backwards: $f(x-h)$ moves the graph right h. Changes *outside* act on y.
- $f(g(x))$ means g first, then f. Its domain is found before simplifying.
- To find an **inverse**, write $y = f(x)$, solve for x, and swap the letters. The domain of f^{-1} is the range of f.

Worked examples

Example 1 · Two restrictions at once

Find the domain of $f(x) = \dfrac{\sqrt{2x+8}}{x-3}$.

Solution. The root needs $2x + 8 \geq 0$, that is $x \geq -4$. The denominator is zero at $x = 3$.

Domain: $x \geq -4$ and $x \neq 3$.

Example 2 · Composition in both orders

Let $f(x) = \dfrac{1}{x}$ and $g(x) = x - 4$. Find $f(g(x))$ and $g(f(x))$ with their domains.

Solution. $f(g(x)) = \dfrac{1}{x-4}$, defined for $x \neq 4$. $g(f(x)) = \dfrac{1}{x} - 4$, defined for $x \neq 0$.

The two compositions are different functions with different domains.

Example 3 · Inverse of a rational function

Find the inverse of $f(x) = \dfrac{x-1}{x+2}$.

Solution. $y(x+2) = x - 1$ gives $xy - x = -2y - 1$, so $x = \dfrac{-2y-1}{y-1} = \dfrac{2y+1}{1-y}$.

$f^{-1}(x) = \dfrac{2x+1}{1-x}$.

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Practice

CORE

1. Find the domain of $f(x) = \sqrt{10 - 5x}$.

2. State the range of $y = |x| - 3$.

3. Let $f(x) = 4 - x^2$ and $g(x) = 2x + 1$. Find $f(g(1))$ and $g(f(1))$.

4. Find the inverse of $f(x) = 5x - 3$.

EXAM LEVEL

5. Find the domain of $f(x) = \ln(x^2 - 4)$.

6. Let $f(x) = \sqrt{x}$ and $g(x) = 9 - x^2$. Find $f(g(x))$ and state its domain.

7. Find the inverse of $f(x) = e^{2x} + 1$, and state the domain of the inverse.

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8. Describe how the graph of $y = \dfrac{3}{x+2} - 1$ is obtained from the graph of $y = \dfrac{1}{x}$, and write equations for its two asymptotes.

Question 9 · Free-response practice

Let $f(x) = x^2 - 6x$ with domain $x \geq 3$.

(a) Find the range of f.

(b) Find $f^{-1}(x)$. (Completing the square is helpful.)

(c) State the domain and range of f^{-1}, and verify that $f^{-1}(f(5)) = 5$.

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Worked solutions

1. $10 - 5x \geq 0$, so $x \leq 2$.
2. $y \geq -3$.
3. $g(1) = 3$ and $f(3) = -5$. $f(1) = 3$ and $g(3) = 7$.
4. $f^{-1}(x) = \dfrac{x+3}{5}$.
5. $x^2 - 4 > 0$, so $x < -2$ or $x > 2$.
6. $\sqrt{9 - x^2}$, with domain $-3 \leq x \leq 3$.
7. $y - 1 = e^{2x}$ gives $x = \dfrac{1}{2}\ln(y-1)$, so $f^{-1}(x) = \dfrac{1}{2}\ln(x-1)$ with domain $x > 1$.
8. Stretched vertically by 3, moved left 2 and down 1. Asymptotes: $x = -2$ and $y = -1$.
9. (a) $f(x) = (x-3)^2 - 9$, which is least at $x = 3$: range $y \geq -9$.
   (b) $y + 9 = (x-3)^2$ with $x \geq 3$ gives $x = 3 + \sqrt{y+9}$, so $f^{-1}(x) = 3 + \sqrt{x+9}$.
   (c) Domain $x \geq -9$, range $y \geq 3$. $f(5) = -5$ and $f^{-1}(-5) = 3 + \sqrt{4} = 5$.

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